
When Are Recursive Language Models Useful? A Cost-Aware, Task-Conditional Evaluation

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Abstract

1 Recursive Language Models (RLMs) handle long inputs by writing programs over
2 externalized context and making additional model calls. For a question asking
3 which label is most frequent for one user, an RLM can infer a program while
4 a route that knows the operation can label and count directly. We ask when
5 the RLM’s added adaptivity justifies its failures, latency and cost. We compare
6 released standard RLM with deterministic operators, retrieval, semantic workers
7 and direct inference under common information, scoring and budget contracts. On
8 controlled operations and Oolong tasks with a known operation family, matched
9 routes generally match or outperform standard RLM at lower cost. A replication on
10 unused Oolong source corpora preserves this direction, although its clustered exact-
11 rate difference includes zero. On open multi-document search, RLM beats BM25
12 on a first untouched 45-query shard, but neither a prospectively locked equal-
13 cap comparison nor a same-query repeat yields an RLM contrast that survives
14 multiplicity correction. Route ordering changes across these rollouts while RLM
15 remains less reliable and 5.8–15.9× costlier than stronger alternatives. Controls
16 separate operation choice, answer formatting and execution failure. Restricting
17 comparison to completed queries can reverse the ordering. We therefore require
18 any deployment advantage to hold against a strong task-matched alternative on all
19 attempted queries.

20 1 Introduction

21 A known aggregation can use a local worker and fixed code; open search instead requires choosing
22 what to retrieve and when to stop. Recursive programs can serve both settings, but capability alone
23 does not determine whether their added adaptivity is worth its failures, latency and cost.

24 RLMs [Zhang et al., 2026] make this distinction consequential: a controller writes Python over
25 externalized context and may recursively query models. Existing studies compare task-agnostic long-
26 context and coding scaffolds, vary depth or training and add self-reflective search, typed combinators
27 and recursive harnesses [Zhang et al., 2026, Bertsch et al., 2026, Alizadeh et al., 2026, Roy et al.,
28 2026, Lumer et al., 2026]. Deployment alternatives include RAG–long-context routing [Li et al.,
29 2024, Chen et al., 2026a], retrieval [Lewis et al., 2020], prompt compression [Jiang et al., 2023],
30 deterministic operators and decomposed workers [Khot et al., 2023]. The relevant comparison is
31 RLM against the strongest affordable route permitted by the task contract.

32 We define a task family by its operation and answer schema. An information contract states the
33 context, tools and prior knowledge available to each route; a budget covers calls or decisions, time
34 and measured cost. We then ask whether adaptive program generation yields a stable accuracy–cost
35 advantage over operation-matched alternatives. This extends cost-aware agent auditing [Kapoor et al.,
36 2025, 2026, Bragg et al., 2026] and differs from RLM trajectory diagnosis: the unit is the deployed

37 route, not recursion depth or one component. Declared specialization, paired uncertainty and rollout
38 stability determine the credibility of the task-conditional route frontier [Yu and Kumbier, 2020].

39 We compare released standard RLM with deterministic, retrieval, semantic-worker and direct routes.
40 Oolong and an RLM trajectory already decompose labeling from aggregation [Zhang et al., 2026,
41 App. E.2]; our study adds a same-row comparison of complete deployed routes. Table 3 summarizes
42 the boundary. We contribute:

- 43 1. An operation-matched protocol that fixes rows and scorers, declares route-specific prior
44 knowledge and reports accuracy, failures, latency, calls and cost. An answer-contract repair
45 separates finalization failure from route quality.
- 46 2. Wrong-family, same-operator, finalization and trace controls that distinguish operation
47 knowledge from tool access and localize differences in semantic judgment, output schema
48 and scaffold reliability; prior RLM trajectory diagnoses do not combine these controls
49 [Zhang et al., 2026, Wang, 2026].
- 50 3. A bounded deployment comparison: known-operation routes lead on the evaluated Oolong
51 slices, while adaptive-search differences remain unresolved. Paired rollout and completion
52 analyses show how the same outputs can support different apparent rankings when the
53 evaluation population changes.

54 2 Operation-Matched Evaluation

55 **Scope.** The audited system is the official `r1ms` v0.1.3 inference-time scaffold: a controller generates
56 Python over externalized context and may call additional models before answering. We compare this
57 implementation with task-matched routes on identical instances. The exact-aggregation, semantic-
58 counting and Oolong tasks have direct standard-RLM comparisons; the classification task is diagnostic.
59 Tested controller changes and official trained-policy/code checks probe robustness, but do not
60 reproduce the original SRLM, λ -RLM or RAH benchmark protocols. Conclusions remain specific to
61 the evaluated implementation and settings.

62 **Tasks and independence.** Table 1 defines three controlled families: exact structured aggregation (T-
63 STRUCT), directional semantic counting (T-MIXED) and standard classification (T-NLP, diagnostic
64 only). Oolong separates local semantic labeling from aggregation. Label-provided slices (LP-80 and
65 LP-150) isolate aggregation and answer formatting rather than end-to-end performance. The standard-
66 unlabeled slice (SU-150) removes those labels but reuses development contexts, so it supplies scale
67 evidence. Two 45-case slices test different limits: the context-disjoint slice (CD-45) excludes prior
68 context windows and spans 1k, 4k and 262k tokens; the validation-source slice (VC-45) uses two
69 previously unused corpora from the official validation split. Oolong supplies no source-document
70 IDs, so neither slice establishes document-level independence.

71 BrowseComp-Plus tests search in fixed pools of 1,000 documents per query [Chen et al., 2026b].
72 The original 80-query study supplies exploratory evidence; a first untouched 45-query shard tests
73 replication. A second untouched 45-query shard adds iterative search and common decision/time
74 caps before scoring. A later seed repeat reuses those same queries and is a rollout sensitivity, not 45
75 new independent cases. LongBench-v2 Academic QA ($N=94$) remains a separate format-validity
76 diagnostic [Bai et al., 2025].

77 **What the typed route does.** Consider an Oolong question asking whether one label is more
78 frequent than another for a named user. The worker labels individual records; fixed code filters that
79 user’s records, counts the two labels and formats their relation. The route knows the operator family
80 but receives no gold local labels. An RLM must instead choose and execute its program. Comparing
81 complete routes therefore tests the value of that additional adaptivity under declared specialization.

82 **Evaluation design.** Figure 1 specifies the information available to each route. A complete route
83 includes input handling, model or tool calls, answer finalization and failures. Task-specific operators
84 and answer schemas are intentional deployment specialization, so comparisons are not pure scaffold
85 effects. Wrong-family, same-tools and finalizer controls test whether an observed advantage comes
86 from operation knowledge, access to tools or answer formatting. Frozen rows, scorer versions,
87 decision/time caps and failure records link each result to its comparison contract.

Table 1: Controlled task-family construction. Each family targets a different long-context operation with a prespecified alternative route.

Family	Operation	N	Matched alternative
T-STRUCT	Exact pair enumeration	30	SLM extraction + fixed operator
T-MIXED	Directional semantic counting	30	Direct LLM / local judgments
T-NLP (diagnostic)	Standard classification	24	SLM direct

Table 2: Model roles and their corresponding evaluation settings.

Role	Model	Use
RLM controller	Gemini 2.5 Flash; GPT-5; Qwen3.6-35B-A3B	Standard scaffold; direct matched Gemini; BrowseComp matched routes
Trained/code RLM	Qwen3-30B-A3B; λ /Qwen3-8B	Frozen CD-15 diagnostics
Direct LLM	Gemini 2.5 Flash; Qwen3.6-35B-A3B	Direct-context and matched-controller baselines
SLM worker	Gemma 4 26B-A4B	Direct, CoT, labeling and fallback routes
Cheap controller	GPT-4o-mini	Deliberately weak cheap controller; T-NLP orchestration only

Baselines and model roles. Table 2 identifies the models. The primary Oolong RLM controller is Gemini 2.5 Flash; T-STRUCT uses GPT-5. All BrowseComp routes use Qwen3.6-35B-A3B. There, the text route issues one batch of textual subqueries, while iterative search updates its lexical query after each retrieved batch; neither generates and executes Python. Within each endpoint, decoding is matched: the first 45-query and equal-cap endpoints use temperature 0.7, seed 20 and thinking disabled; the repeat changes only the seed to 21. The original 80-query endpoint and both judges use temperature 0. Self-hosted models are pinned by revision and container digest. Appendix H gives the full settings, hosted-model identifiers and protocol differences.

Evidence roles. LP isolates aggregation; SU tests end-to-end scale on reused contexts; VC tests unused-source transfer; and CD tests specialization and shared tools. For search, the prospectively locked equal-cap endpoint is primary, the two earlier studies provide context and the same-query seed repeat is a rollout sensitivity. These roles are not pooled.

Estimands, scoring and uncertainty. For paired routes r, s on n rows, let Y_{ir} be exact correctness and C_{ir} measured cost for row i under route r . We estimate $\hat{\Delta}_A = n^{-1} \sum_i (Y_{ir} - Y_{is})$ and $\hat{\Delta}_C = n^{-1} \sum_i (C_{ir} - C_{is})$; failed or invalid executions have $Y = 0$. A route is empirically dominated if another has no lower observed accuracy and lower cost. This point-estimate relation does not resolve uncertainty or supply a deployment-specific value for accuracy.

Paired accuracy comparisons use exact McNemar tests. For routes reported in order, b counts cases correct only for the first and c cases correct only for the second. BrowseComp difference intervals use 50,000 paired-row bootstrap draws. Oolong intervals resample context-window clusters (5,000 draws): 72 for development $N=120$, 73 for SU-150, 32 for CD-45 and 20 for VC-45. Intervals are conditional on observed executions. Within each equal-cap rollout, Holm correction covers the three RLM comparisons against BM25, text and iterative search. Subgroup, controller and mechanism probes are descriptive; there is no global familywise claim.

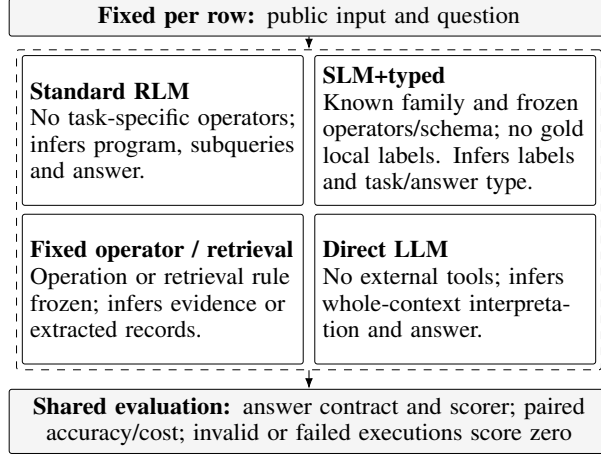


Figure 1: Route and information contracts. Allowed deployment prior knowledge is declared before comparison. The estimates compare complete deployed routes. They do not isolate a scaffold-only causal effect or evaluate an automatic router.

Oolong reports binary exact accuracy and the benchmark’s partial-credit mean score: numeric near-misses receive $0.75^{|\Delta|}$ and relative-frequency answers accept the prescribed relation substrings. BrowseComp uses a method-blinded Gemma 4 semantic judge and a separate GPT-5.6 audit. Both compare predictions with benchmark gold answers; cross-judge agreement measures stability, not human validity. MCQ diagnostics map predictions back after answer shuffling.

Cost accounting. RLM cost is decomposed into controller generation, child-query execution and final extraction. API totals capture successful completions and are lower bounds where failed calls return no usage. Direct and SLM baselines are single-call unless retrieval-augmented; fixed operators have zero marginal model cost but nonzero engineering cost. Engineering amortization is an explicitly hypothetical sensitivity (Appendix A), not measured labor or total cost of ownership. Self-hosted BrowseComp costs use active runtime by route with generation-plus-judge total reported separately. Reported statistics and cost totals are computed from stored logs.

3 Results

3.1 Known-operation tasks

Matched routes lead on controlled tasks. On T-STRUCT, Gemma extracts records and a fixed Python operator performs the aggregation, which is exact by construction, so this row measures extraction success and cost. It reaches 100% accuracy versus 96.7% for RLM at about $99\times$ lower captured deployed-configuration API cost; because both are near ceiling, the result establishes cost dominance only, and supports neither a reliable accuracy difference nor a pure matched-model scaffold-overhead estimate. T-MIXED requires semantic judgments. A matched Gemini ablation separates finalization from semantic error: the count-only finalizer returns no valid answers while the JSON finalizer returns valid answers on all 20 examples and solves 13. A frozen $N=30$ endpoint compares direct Gemini with JSON-finalized RLM: direct Gemini solves 28/30 at \$0.0223, RLM solves 18/30 at \$0.1089 (paired exact $b=12$, $c=2$, $p=0.0129$) and remains about $4.9\times$ costlier. We treat T-NLP only as an interface diagnostic because no official RLM accuracy run is logged; moreover, the classification target requires categorical labels while the current aggregation scaffold assumes numeric-style answers.

Oolong specialization persists across labels and sources. On label-provided LP-80, RLM reaches 82.4% mean score against 70.0% for a generic fixed-parser/SLM fallback, although that difference is uncertain ($p=0.089$); its largest gain is on timeline questions and deterministic parsing remains perfect on counting. Gold-label typed operators reach 100% on the same split, so RLM is not the strongest LP-80 route. On frozen held-out LP-150, RLM reaches 84.7% exact while typed

Table 3: Main route results and boundaries. LP-80 is mean score; LP-150 and all other accuracy entries are exact. “Best non-RLM” prioritizes accuracy on T-MIXED. T-STRUCT compares deployed configurations; CD-45 is diagnostic. Equal-cap BrowseComp active-runtime cost is \$13.279 (\$15.015 including lower-bound interruption billing and judging).

Role / regime	Best non-RLM	Standard RLM	Boundary supported by this row
T-STRUCT controlled aggregation	Gemma extract + fixed agg.: 100%, \$0.0002/ex	96.7%, \$0.0198/ex	Near-ceiling accuracy; about 99× lower captured cost in this cross-model comparison. This does not isolate scaffold overhead.
T-MIXED controlled count	Direct Gemini: 28/30, \$0.0223 total	RLM: 18/30, \$0.1089 total	Same-model direct inference wins on the frozen $N=30$ endpoint; the $N=20$ finalizer ablation isolates answer-contract failure.
Oolong CD-45 mechanism controls	Correct-family SLM+typed: 36/45; wrong-family: 2/45	Same-tools RLM: 31/45 or 29/45; GPT-5/GPT-5.6 Luna: 23/45, 26/45; at least 2.3× costlier	Choosing the operator family matters here; shared tools and tested controllers do not reverse the ordering.
Oolong LP-80/LP-150 aggregation-only	Typed ops: 100% / 97.3%	RLM: 82.4% / 84.7%	Provided labels isolate aggregation and answer formatting; these are not end-to-end results.
Oolong SU-150/VC-45 end-to-end	SLM-labeler + typed ops: 98/150, \$0.839; unused-source VC-45 35/45, \$0.312	RLM 68/150, \$4.145; VC-45 28/45, \$1.139	SU-150 reuses contexts. VC-45 tests new sources: its exact-rate difference includes zero, while its mean-score difference does not.
BrowseComp equal-cap primary ($N=45$)	Text: 37/45, \$1.400; iterative: 33/45, \$1.710; BM25: 25/45, \$0.296	RLM: 31/45, \$9.873	All three RLM comparisons remain unresolved after Holm correction; RLM costs 5.8–7.1× more than iterative/text routes.

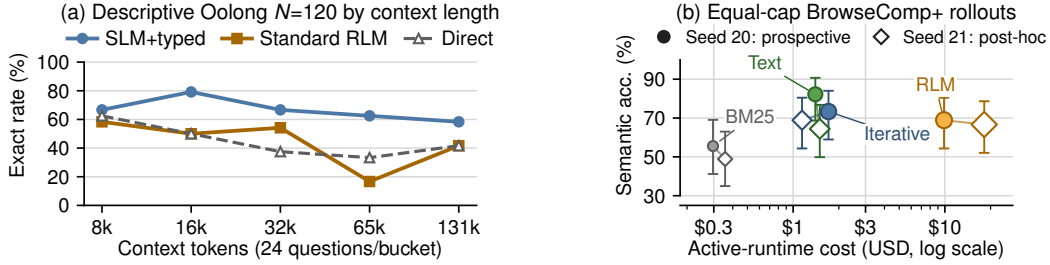


Figure 2: Accuracy and cost across task regimes. (a) Descriptive exact rates on standard-input Oolong development data ($N=120$; 24 per bucket). The inferential contrast is the paired cluster-resampled difference reported in the text, not a per-cell interval. (b) Accuracy versus active-runtime cost on the matched-Qwen equal-cap endpoint (filled circles: prospective seed 20; hollow diamonds: same-query seed 21; $N=45$ each). Whiskers are Wilson intervals, and larger markers mean more all-row model calls (RLM/iterative/text/BM25: 476/426/311/42 and 667/276/312/42). No RLM contrast survives Holm correction in either rollout.

144 rules fixed before evaluation reach 97.3% ($b=23, c=4, p<0.001$). Gold local labels make these
145 aggregation/output-schema ablations, not end-to-end Oolong results. They show that RLM handles
146 flexible schemas better than a generic parser but not that open-ended Python is uniquely required
147 when the schema family is known.

148 All central standard-unlabeled checks use `context_window_text` without gold local labels. An
149 $N=120$ development replication established the SLM>RLM direction across every 8k–131k bucket
150 (Figure 2(a)), with a paired 22.5-point exact-rate advantage ($b=37, c=10, p=9.8 \times 10^{-5}$; 72-cluster
151 interval 11.9–33.0). The fixed pipeline was evaluated on SU-150, CD-45 and VC-45, which was
152 frozen before model outputs were generated. Before scoring, one parser repair was applied only to
153 rows returning no call, a condition visible without labels.

154 On SU-150, SLM+typed solves 98/150 at \$0.839, versus RLM 68/150 at \$4.145 and direct Gemini
155 65/150 at \$1.939. Discordance favors SLM over RLM ($b=44, c=14, p=1.00 \times 10^{-4}$); no direct–RLM

156 difference is detected ($p=0.788$). The 73-cluster interval gives a 10.6–29.9 point SLM advantage.
157 Because all rows reuse previously evaluated context windows, SU-150 shows that the locked method
158 retains its advantage at larger N but is not an independence check.

159 VC-45 removes the known source-corpus overlap: its official validation rows come from spam
160 and `trec_coarse`, absent from the test split and all previous Oolong evaluations. A replay of
161 the unchanged question parser agrees with the recorded task and answer-type metadata on all 150
162 SU-150, 45 CD-45 and 45 VC-45 rows. This validates the parser on these benchmark templates, not
163 on arbitrary paraphrases or unseen operator families. SLM+typed solves 35/45 at \$0.312, versus RLM
164 28/45 at \$1.139 and direct Gemini 23/45 at \$0.658. SLM–RLM discordance is $b=10$, $c=3$, $p=0.092$
165 (Holm $p=0.185$); the 20-cluster exact-difference interval is 0.0–31.1 points while the mean-score
166 interval is 3.0–30.9. RLM reaches 13/15 on user questions but 6/15 on counting. Task and source
167 are not factorially balanced, so subgroup differences are descriptive. The specialized route remains
168 cheaper and directionally favored under source-corpus transfer but VC-45 alone establishes neither a
169 significant exact-rate advantage nor document-level independence.

170 CD-45 retains the specialization controls: using the wrong operator family collapses the typed route
171 from 36/45 to 2/45. Giving standard RLM the same operator library yields 31/45 when the library
172 includes a high-level convenience solver and 29/45 with only lower-level atomic tools, but remains
173 at least $2.3\times$ costlier. An independent standard-RLM repeat drops from 27/45 to 21/45. Together,
174 SU-150 supplies scale, VC-45 supplies source-corpus transfer and CD-45 isolates specialization and
175 same-tools behavior; none is a full Oolong leaderboard.

176 **Tested controllers preserve the Oolong ordering.** Strict-stop GPT-5 and GPT-5.6 Luna standard-
177 RLM runs solve 23/45 and 26/45 CD-45 cases, below SLM+typed’s 36/45 (descriptive paired
178 $p=0.00235$ and $p=0.041$). On CD-15, the official trained Qwen3-30B-A3B policy solves 10/15 and
179 unmodified public λ -RLM code with Qwen3-8B solves 8/15, versus 15/15 for SLM+typed. The λ run
180 counts two 1,800-second timeouts incorrect. These sensitivities do not reverse the observed ordering,
181 but differing populations and fidelity tiers preclude pooling. They support a limited robustness claim,
182 not full official-family benchmark parity (Appendix A).

183 3.2 Adaptive search

184 **Stronger alternatives leave the comparison unresolved.** BrowseComp-Plus supports controlled
185 retrieval/agent comparisons. We separate three stages: an exploratory 80-query study, a first untouched
186 45-query replication and the prospective equal-cap comparison below. The exploratory study gave
187 RLM/text/BM25 54/48/39 correct, but completion across two episodes after partial score visibility
188 limits its evidential role. On the first untouched shard, the same routes scored 33/29/21. RLM
189 exceeded BM25 by 26.7 points (paired 95% interval 11.1–44.4; Holm $p=0.0151$), while its 8.9-point
190 lead over text remained uncertain (−6.7–24.4). RLM completed 42/45, cost \$6.746 and had a
191 425.7-second p95 latency; text completed 45/45 at \$1.610 and 44.9 seconds. A blinded second-model
192 audit agreed on 128/129 successful outputs ($\kappa=0.983$). The positive RLM–BM25 result is retained;
193 it does not establish superiority to adaptive search.

194 The prospective equal-cap endpoint tests four routes on a second untouched 45-query shard. We added
195 iterative lexical search and attempted all 180 Qwen3.6 route cells under common 20-decision/1,200-
196 second caps before semantic scoring. RLM scores 31/45 (68.9%), versus iterative 33/45 (73.3%),
197 text 37/45 (82.2%) and BM25 25/45 (55.6%). Differences are −4.4, −13.3 and +13.3 points,
198 with discordances 6/8, 5/11 and 10/4. Paired 95% intervals are [−20.0, 11.1], [−31.1, 4.4] and
199 [−2.2, 28.9] points; no contrast survives Holm correction ($p_{\text{Holm}} \geq 0.539$). The second judge closely
200 agrees ($\kappa=0.985$).

201 RLM completes 41/45 with 50.1/628.8-second median/p95 latency. Iterative/text complete all 45
202 with 14.1/90.2 and 22.1/47.6 seconds; BM25 completes 42/45. On 39 common completions, scores
203 are 30/28/31/22; this post-treatment sensitivity leaves the adaptive routes near parity. A question-only
204 Qwen leakage probe scores 0/45 under both judges: neither finds a correct answer without the corpus.
205 This endpoint does not establish the earlier RLM–BM25 advantage; because both queries and the
206 resource contract changed, the difference cannot be attributed to equal caps alone; Figure 2(b) plots
207 both equal-cap rollouts against active-runtime cost. Open search alone is not a sufficient routing
208 signal.

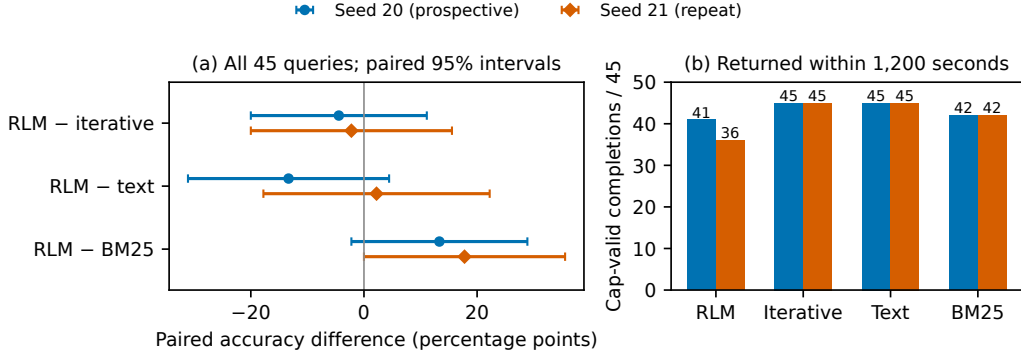


Figure 3: Paired accuracy and completion on the same 45 search queries. (a) RLM minus each alternative, with paired 95% bootstrap intervals conditional on each rollout; these marginal contrast intervals are not simultaneous confidence bands. Holm-adjusted tests resolve no RLM advantage. (b) Outputs returned within the 1,200-second cap. The repeat has 37 successful RLM executions but only 36 cap-valid completions. Dropping incomplete cases changes the comparison population.

Rollouts and completion subsets answer different questions. A post-hoc seed-21 repeat also leaves the RLM comparisons unresolved. RLM/iterative/text/BM25 complete 37/45, 45/45, 45/45 and 42/45 and score 30/31/29/22; no RLM contrast survives Holm ($p_{\text{Holm}} \geq 0.346$). A cap-valid completion finishes within 1,200 seconds: RLM has 36/45, and on 34 rows cap-valid for every route the scores are 28/23/20/18. This post-treatment subset puts RLM first, but selecting cases by completion changes the population; it does not identify a causal effect of attrition. RLM costs 12.1–15.9 $\times$ the adaptive routes. Sol agrees on 166/169 outputs ($\kappa=0.961$); the repeat is not pooled with the prospective endpoint.

On seed 21, RLM trails iterative search by 2.2 points over all 45 queries but leads by 14.7 points on the 34 common cap-valid completions: a 16.9-point change from selection alone. For text, the corresponding differences are +2.2 and +23.5 points. A paired comparison of the RLM–text difference across seeds changes by 15.6 points, with a case-bootstrap interval of $[-8.9, 40.0]$. This descriptive interval is conditional on the two observed rollouts; it does not establish a population-level seed effect. Figure 3 displays the all-case comparisons and their completion denominators.

Implications for route choice. Known operation families warrant fixed-operator and semantic-worker baselines. Open search warrants strong retrieval and adaptive nonrecursive alternatives. Our results support these comparisons, not an automatic router: flexible schemas do not uniquely require generated Python, and open search alone does not establish an RLM advantage.

4 Mechanism Analysis

Failures occur in task interpretation, evidence selection, aggregation, tool use and finalization rather than one RLM component. Appendix C.1 groups these observations by known-operator overhead, answer-contract sensitivity, semantic-local error, output-schema rigidity, search instability, context constraints and multiple-choice bias.

Additional orchestration need not help. A strong LLM can delegate local interpretation to cheaper workers [Narayan et al., 2025], but this adds little when the operation is already fixed. In the T-NLP diagnostic, LLM+SLM scores 83.3% versus 87.5% for SLM direct at about 3.4 $\times$ the cost. This supports checking a direct worker before adding orchestration; it is not an official RLM comparison.

Traces expose failures at distinct stages. CD-45 case 710070026 illustrates a typed-route failure: all seven local labels are correct, yet the route returns “contradiction” rather than the gold least-frequent label “neutral”; RLM and direct inference are correct. Thus accurate local labeling does not guarantee correct aggregation or tie handling. Conversely, case 312030020 is solved by RLM and the typed route while direct inference returns the wrong user. These outcome-selected examples

241 explain distinct failure stages, not their prevalence (Appendix C.2). Stored T-MIXED traces likewise
242 show successful dual-gate filters but also vocabulary overreach and brittle finalization. These traces
243 localize where failures surface; they do not attribute them to the scaffold as a whole.

244 **Multiple-choice position bias limits the QA diagnostic.** LongBench-v2 Academic QA is multiple-
245 choice so answer-position priors confound retrieval-vs.-RLM comparisons. On the $N=50$ discovery
246 set, canonical BM25 predicts A 68% of the time while gold-A is 18%. Rotating priming improves
247 pooled $N=94$ ($p=0.015$) but not held-out $N=44$ ($p=1.0$); single-call accuracy is unmeasured. We
248 therefore keep it as a validity diagnostic, with prior debiasing comparisons in Appendix B.

249 5 Related Work

250 RLM externalizes context through controller-generated programs and recursive calls [Zhang et al.,
251 2026]. Follow-ups test whether recursion drives gains or add typed operators, coding agents and
252 recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Ehrlich and Blackman, 2026, Cao et al.,
253 2026, Lumer et al., 2026]; Wang reports task-dependent depth, cost and strong-model degradation
254 [Wang, 2026]. Unlike trajectory studies, our unit is the complete route, compared on matched rows
255 and scorers with paired uncertainty.

256 Cost-aware agent evaluation measures answers and the resources used to obtain them [Kapoor et al.,
257 2025, 2026, Bragg et al., 2026]; RAG inference scaling allocates a given input-token budget [Yue
258 et al., 2025]. We apply both ideas to complete long-context routes under explicit information, scoring
259 and budget contracts and test recommendation stability across rollouts.

260 Oolong separates local labeling from aggregation, which supports our mechanism ablations [Bertsch
261 et al., 2026]. Long-context suites test reasoning, effective context and retrieval/length effects [Bai
262 et al., 2025, Chen et al., 2026c, Hsieh et al., 2024, Yen et al., 2025, Zhang et al., 2024, Modarressi
263 et al., 2025, Li et al., 2025]. Decomposition, search and routing systems select evidence or execution
264 routes [Xu et al., 2026, Song et al., 2026, Ong et al., 2025, Li et al., 2024, Chen et al., 2026a];
265 search-space scale can also change lexical-versus-agentic ordering [Wang et al., 2026]. Our claims
266 remain limited to the declared fixed pools.

267 6 Limitations

268 **External validity.** We diagnose task-conditional failure modes, not average long-context perfor-
269 mance. VC-45 transfers source corpora but not documents because Oolong lacks IDs; SU-150 reuses
270 windows. Official-family evidence covers CD-15 and the controlled curve ends at 131k; longer rows
271 are stress tests.

272 **Measurement.** The 180 equal-cap cells run under fixed caps across three execution episodes. Cross-
273 family judge agreement and a question-only leakage control assess stability, not human validity; there
274 is no human adjudication amid known contextual judge fragility [Tan et al., 2025, Xu et al., 2025].

275 **Cost and scope.** Costs exclude engineering and TCO; alternatives are not exhaustive and recursion
276 depth is fixed. BrowseComp bins as small as $n=10$ preclude subgroup claims. The below-RLM
277 same-row embedding rerank is not full-corpus dense retrieval (Appendix B). Seven repeat RLM calls
278 exceed the cap and score incorrect; self-hosted and API dollars remain separate. Automated routing
279 could suppress useful methods in untested domains.

280 **Ethics and AI assistance.** Public and synthetic benchmarks involve no human subjects. AI tools
281 assisted coding, experiment orchestration and editing; the authors verified all outputs.

282 7 Conclusion

283 Specialized routes lead on tested known operations at lower cost, while equal-cap search yields
284 no RLM advantage despite $5.8\text{--}15.9\times$ higher cost. Completion filtering can change the winner,
285 so recursion should be judged against operation-matched alternatives using all-query accuracy,
286 completion and cost.

References

- Keivan Alizadeh, Parshin Shojaei, Minsik Cho, and Mehrdad Farajtabar. Recursive language models meet uncertainty: The surprising effectiveness of self-reflective program search for long context. *arXiv preprint arXiv:2603.15653*, 2026. URL <https://arxiv.org/abs/2603.15653>.
- Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench v2: Towards deeper understanding and reasoning on realistic long-context multitasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, pages 3639–3664, 2025. URL <https://arxiv.org/abs/2412.15204>.
- Amanda Bertsch, Adithya Pratapa, Teruko Mitamura, Graham Neubig, and Matthew R. Gormley. Oolong: Evaluating long context reasoning and aggregation capabilities. In *Conference on Language Modeling*, 2026. URL <https://colm.eventhosts.cc/Conferences/2026/AcceptedPapers>.
- Jonathan Bragg, Mike D’Arcy, Nishant Balepur, Dan Bareket, Bhavana Dalvi Mishra, Sergey Feldman, Dany Haddad, Jena D. Hwang, Peter Jansen, Varsha Kishore, Bodhisattwa Prasad Majumder, Aakanksha Naik, Sigal Rahamimov, Kyle Richardson, Amanpreet Singh, Harshit Surana, Aryeh Tiktinsky, Rosni Vasu, Guy Wiener, Chloe Anastasiades, Stefanus Candra, Jason Dunkelberger, Daniel Emery, Rob Evans, Malachi Hamada, Regan Huff, Rodney Kinney, Matt Latzke, Jaron Lochner, Ruben Lozano-Aguilera, Ngoc-Uyen Nguyen, Smita Rao, Amber Tanaka, Brooke Vlahos, Peter Clark, Doug Downey, Yoav Goldberg, Ashish Sabharwal, and Daniel S. Weld. AstaBench: Rigorous benchmarking of AI agents with a scientific research suite. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=M7TNf5J26u>.
- Weili Cao, Xunjian Yin, Bhuwan Dhingra, and Shuyan Zhou. Coding agents are effective long-context processors. *arXiv preprint arXiv:2603.20432*, 2026. URL <https://arxiv.org/abs/2603.20432>.
- Yiwen Chen, Kuan Li, Fuzhen Zhuang, Deqing Wang, Zhao Zhang, Liwen Zhang, Yong Jiang, Shuai Wang, and Minhao Cheng. Route before retrieve: Activating latent routing abilities of LLMs for RAG vs. long-context selection. *arXiv preprint arXiv:2605.10235*, 2026a. URL <https://arxiv.org/abs/2605.10235>.
- Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Sahel Sharifymoghaddam, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Yanxi Li, Haoran Hong, Xinyu Shi, Xuye Liu, Hosna Oyarhoseini, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhui Chen, and Jimmy Lin. BrowseComp-Plus: A fair and disentangled evaluation benchmark for deep search agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026b. URL <https://aclanthology.org/2026.acl-long.1023/>.
- Ziyang Chen, Xing Wu, Junlong Jia, Chaochen Gao, Qi Fu, Debing Zhang, and Songlin Hu. LongBench pro: A more realistic and comprehensive bilingual long-context evaluation benchmark. *arXiv preprint arXiv:2601.02872*, 2026c. URL <https://arxiv.org/abs/2601.02872>.
- Clint Ehrlich and Theodore Blackman. LCM: Lossless context management. *arXiv preprint arXiv:2605.04050*, 2026. URL <https://arxiv.org/abs/2605.04050>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? In *Proceedings of the First Conference on Language Modeling (COLM 2024)*, 2024. URL <https://arxiv.org/abs/2404.06654>.
- Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. LLMingua: Compressing prompts for accelerated inference of large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, 2023. URL <https://arxiv.org/abs/2310.05736>.
- Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. AI agents that matter. *Transactions on Machine Learning Research*, 2025-May, 2025. URL <https://arxiv.org/abs/2407.01502>.

337 Sayash Kapoor, Benedikt Stroebl, Peter Kirgis, Nitya Nadgir, Zachary Siegel, Boyi Wei, Tianci
338 Xue, Ziru Chen, Felix Chen, Saiteja Utpala, Franck Ndzomga, Dheeraj Oruganty, Sophie Luskin,
339 Kangheng Liu, Botao Yu, Amit Arora, Dongyoon Hahm, Harsh Trivedi, Huan Sun, Juyong
340 Lee, Tengjun Jin, Yifan Mai, Yifei Zhou, Yuxuan Zhu, Rishi Bommasani, Daniel Kang, Dawn
341 Song, Peter Henderson, Yu Su, Percy Liang, and Arvind Narayanan. Holistic agent leaderboard:
342 The missing infrastructure for AI agent evaluation. In *International Conference on Learning
343 Representations*, 2026. URL <https://openreview.net/forum?id=vUaY1t64ZZ>.

344 Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish
345 Sabharwal. Decomposed prompting: A modular approach for solving complex tasks. In *Pro-
346 ceedings of the International Conference on Learning Representations (ICLR 2023)*, 2023. URL
347 <https://arxiv.org/abs/2210.02406>.

348 Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal,
349 Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe
350 Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural
351 Information Processing Systems (NeurIPS 2020)*, volume 33, pages 9459–9474, 2020. URL
352 <https://arxiv.org/abs/2005.11401>.

353 Kuan Li, Liwen Zhang, Yong Jiang, Pengjun Xie, Fei Huang, Shuai Wang, and Minhao Cheng.
354 LaRA: Benchmarking retrieval-augmented generation and long-context LLMs – no silver bullet for
355 LC or RAG routing. In *Proceedings of the 42nd International Conference on Machine Learning*,
356 volume 267 of *Proceedings of Machine Learning Research*, pages 36846–36867. PMLR, 2025.
357 URL <https://proceedings.mlr.press/v267/li25dv.html>.

358 Zhuowan Li, Cheng Li, Mingyang Zhang, Qiaozhu Mei, and Michael Bendersky. Retrieval augmented
359 generation or long-context LLMs? a comprehensive study and hybrid approach. In *Proceedings
360 of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*,
361 2024. URL <https://aclanthology.org/2024.emnlp-industry.66/>.

362 Elias Lumer, Sahil Sen, Kevin Paul, and Vamse Kumar Subbiah. Recursive agent harnesses. *arXiv
363 preprint arXiv:2606.13643*, 2026. URL <https://arxiv.org/abs/2606.13643>.

364 Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, Trung Bui, Ryan A. Rossi, Seunghyun
365 Yoon, and Hinrich Schütze. NoLiMa: Long-context evaluation beyond literal matching. In
366 *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL <https://arxiv.org/abs/2502.05167>.

368 Arvind Narayan, Dan Biderman, Sabri Eyuboglu, Avner May, Scott Linderman, James Zou, and
369 Christopher Ré. Minions: Cost-efficient collaboration between on-device and cloud language
370 models. *arXiv preprint arXiv:2502.15964*, 2025. URL <https://arxiv.org/abs/2502.15964>.

371 Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez,
372 M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data.
373 In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2406.18665>.

375 Amartya Roy, Rasul Tutunov, Xiaotong Ji, Matthieu Zimmer, and Haitham Bou-Ammar. The Y-
376 combinator for LLMs: Solving long-context rot with λ -calculus. *arXiv preprint arXiv:2603.20105*,
377 2026. URL <https://arxiv.org/abs/2603.20105>.

378 Maojia Song, Renhang Liu, Xinyu Wang, Yong Jiang, Pengjun Xie, Fei Huang, Soujanya Poria, and
379 Jingren Zhou. Demystifying deep search: A holistic evaluation with hint-free multi-hop questions
380 and factorised metrics. In *The Fourteenth International Conference on Learning Representations*,
381 2026. URL <https://openreview.net/forum?id=x4zQDewgHr>.

382 Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Yuan Tang, Alejandro Cuadron, Chenguang
383 Wang, Raluca Popa, and Ion Stoica. JudgeBench: A benchmark for evaluating LLM-based
384 judges. In *The Thirteenth International Conference on Learning Representations*, 2025. URL
385 <https://openreview.net/forum?id=G0dksFayVq>.

386 Daren Wang. Think, but don’t overthink: Reproducing recursive language models. *arXiv preprint
387 arXiv:2603.02615*, 2026. URL <https://arxiv.org/abs/2603.02615>.

- 388 Pengyu Wang, Benfeng Xu, Shaohan Wang, Mingxuan Du, Xin Zeng, Huarui Wu, Lei Zhang, and
 389 Licheng Zhang. BM25 wins at scale: A scaling study of retrieval-augmented generation paradigms.
 390 *arXiv preprint arXiv:2607.26497*, 2026. URL <https://arxiv.org/abs/2607.26497>.
- 391 Austin Xu, Srikanth Bansal, Yifei Ming, Semih Yavuz, and Shafiq Joty. Does context matter? Contextu-
 392 alJudgeBench for evaluating LLM-based judges in contextual settings. In *Proceedings of the 63rd*
 393 *Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
 394 9541–9564. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.470.
 395 URL <https://aclanthology.org/2025.acl-long.470/>.
- 396 Zhen Xu, Shang Zhu, Jue Wang, Junlin Wang, Ben Athiwaratkun, Chi Wang, James Zou, and
 397 Ce Zhang. When does divide and conquer work for long context LLM? a noise decomposition
 398 framework. In *International Conference on Learning Representations*, 2026. URL [https://](https://openreview.net/forum?id=ddQFUuHDDt)
 399 openreview.net/forum?id=ddQFUuHDDt.
- 400 Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat,
 401 and Danqi Chen. HELMET: How to evaluate long-context language models effectively and
 402 thoroughly. In *Proceedings of ICLR 2025*, 2025. URL <https://arxiv.org/abs/2410.02694>.
- 403 Bin Yu and Karl Kumbier. Veridical data science. *Proceedings of the National Academy of Sciences*,
 404 117(8):3920–3929, 2020. doi: 10.1073/pnas.1901326117. URL [https://doi.org/10.1073/pnas.](https://doi.org/10.1073/pnas.1901326117)
 405 [1901326117](https://doi.org/10.1073/pnas.1901326117).
- 406 Zhenrui Yue, Honglei Zhuang, Aijun Bai, Kai Hui, Rolf Jagerman, Hansi Zeng, Zhen Qin, Dong
 407 Wang, Xuanhui Wang, and Michael Bendersky. Inference scaling for long-context retrieval
 408 augmented generation. In *International Conference on Learning Representations*, 2025. URL
 409 <https://openreview.net/forum?id=FSjIrOm1vz>.
- 410 Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive language models. In *Advances in Neural*
 411 *Information Processing Systems*, 2026. URL <https://arxiv.org/abs/2512.24601>.
- 412 Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Khai Hao, Xu Han,
 413 Zhen Leng Thai, Shuo Wang, Zhiyuan Liu, and Maosong Sun. ∞ Bench: Extending long context
 414 evaluation beyond 100k tokens. In *Proceedings of the 62nd Annual Meeting of the Association for*
 415 *Computational Linguistics (ACL 2024)*, 2024. URL <https://arxiv.org/abs/2402.13718>.

416 A Protocol Details

417 **Cost-aware audit protocol.** Table 4 summarizes the protocol used throughout this study. Because
 418 the audited object is a scaffolded inference procedure, a row should not be promoted from diagnostic
 419 evidence to method evidence unless the input contract, output schema, scorer, failure accounting and
 420 cost trace are matched.

421 **Role-based model coverage.** The primary panel assigns models to deployment roles rather than
 422 constructing a broad leaderboard: Gemini 2.5 Flash controls the logged standard RLM and supplies
 423 direct frontier inference, Gemma 4 26B-A4B supplies SLM worker paths and GPT-4o-mini controls
 424 the short T-NLP LLM+SLM route. GPT-5 is used for the T-STRUCT frontier-controller exception
 425 and a CD-45 controller sensitivity. Many central alternatives are deterministic or structural so the
 426 unresolved breadth question is same-contract coverage of stronger RLM-family systems rather than
 427 additional direct-LLM columns.

428 That stronger-family evidence includes frozen-manifest adapter pressure tests, matched standard-
 429 unlabeled smokes, the CD-15 repeatability study and CD-45 same-operator-library diagnostics.
 430 The unmodified public λ -RLM repository attempts all 15 frozen CD-15 IDs with its published
 431 Qwen3-8B settings and reaches 8/15 exact with two declared timeouts. MIT OASYS’s official
 432 `r1m-qwen3-30b-a3b-v0.1` post-trained policy reaches 10/15 exact with one timeout through the
 433 canonical `r1m` harness and model-card Oolong settings. The λ row is official-code cross-task evidence
 434 rather than an original-paper reproduction; rankable official SRLM and same-contract RAH rows
 435 remain unavailable.